INCLASS/CASE STUDY 3

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Project Report: Integrated Medical Information System (EMR)

Istanbul City Hospital — Landscape WBS & Activity-Level Cost Estimation

1. Work Breakdown Structure (WBS) - Landscape Table

The table below presents the hierarchical project breakdown in horizontal (landscape) format. It maps WBS Code, Level, Element Name, Key Deliverables, Primary Responsibility, and Estimated Duration across 9.6 inches of horizontal width.

WBS Code	Level	WBS Element / Name	Key Deliverables	Primary Responsibility	Duration
1.0	**Level 0**	**EMR System Program**	**Unified Hospital Information System**	**Project Steering Committee**	**12 Mos**
1.1	**Level 1**	**Project Initiation & Architecture**	**Architecture design & compliance approval**	**Project Manager / Architect**	**2 Mos**
1.1.1	Level 2	Stakeholder Requirements	Functional & clinical specs baseline	Lead Business Analyst	4 Wks
1.1.2	Level 2	Security & Regulatory Framework	KVKK / GDPR & 2FA security matrix	Compliance & Security Officer	4 Wks
1.2	**Level 1**	**Hardware & Infrastructure Setup**	**Cloud datacenter & endpoint hardware fleet**	**Infrastructure Team Lead**	**3 Mos**
1.2.1	Level 2	Local Cloud Datacenter Setup	5 server nodes operational in local cloud	DevOps / Infrastructure Eng.	6 Wks
1.2.1.1	Level 3	• Server Rack & Storage Mounting	Configured storage cluster & network nodes	Systems Administrator	2 Wks
1.2.2	Level 2	End-Point Fleet Procurement	4,000 pre-configured medical tablets	Hardware Procurement Lead	8 Wks
1.2.2.1	Level 3	• Tablet Provisioning & MDM Sync	MDM policy & security locks loaded on tablets	IT Support / Operations	3 Wks
1.3	**Level 1**	**Core Software Development**	**Central EMR, mobile app & offline engine**	**Software Engineering Manager**	**6 Mos**

WBS Code	Level	WBS Element / Name	Key Deliverables	Primary Responsibility	Duration
1.3.1	Level 2	Central Database & Backend	High-availability relational DB & REST APIs	Lead Database Developer	12 Wks
1.3.2	Level 2	Web & Mobile Applications	Cross-ward UI, clinician portal (<3 clicks)	Senior Frontend Developer	14 Wks
1.3.3	Level 2	Offline Sync Engine Module	Local sync protocol for low-Wi-Fi areas	Mobile App Specialist	8 Wks
1.3.4	Level 2	Patient Portal (e-Devlet)	Public portal with e-Devlet single sign-on	Full-Stack Developer	6 Wks
1.4	**Level 1**	**System & API Integrations**	**Interoperable e-Nabız, LIS & PACS interfaces**	**Integration Team Lead**	**4 Mos**
1.4.1	Level 2	National Health Integration	e-Nabız API & HL7 messaging channels	Integration Engineer	10 Wks
1.4.2	Level 2	PACS Imaging & LIS Integration	DICOM image viewer & lab result auto-route	Healthcare Systems Specialist	8 Wks
1.5	**Level 1**	**Data Migration & QA**	**Clean historical data & verified security**	**QA & Data Manager**	**3 Mos**
1.5.1	Level 2	Paper & Legacy Data Migration	Digitized paper logs & legacy database transfer	Data Migration Team	8 Wks
1.5.2	Level 2	Security & Penetration Testing	Audit sign-off & vulnerability test reports	External Security Auditor	4 Wks
1.6	**Level 1**	**Deployment & Staff Training**	**Trained medical staff & 24/7 go-live support**	**Change Management Lead**	**2 Mos**
1.6.1	Level 2	On-Site Clinical Training	200 training sessions for doctors & nurses	Clinical Trainer Lead	6 Wks

WBS Code	Level	WBS Element / Name	Key Deliverables	Primary Responsibility	Duration
1.6.2	Level 2	Go-Live Rollout & Hypercare	99.9% uptime launch & 24/7 helpdesk active	Support & Operations Lead	4 Wks

2. Activity-Level Cost Estimation - Landscape Tables

2.1 Resource Requirements & Direct Cost Breakdown

WBS Task Name	Resource Type Required	Quantity / Duration	Unit Rate (TRY)	Total Base Direct Cost (TRY)
1.2.1 Local Cloud Datacenter	Hardware / Cloud Servers	5 Server Nodes (1 Month)	3,000,000 / node	15,000,000
1.2.2 Tablet Fleet Procurement	Hardware (Medical Tablets)	4,000 Units	22,500 / unit	90,000,000
1.3.1 Central Database Dev	Dev Team (Man-hours)	10,000 Hours	2,000 / hr	20,000,000
1.3.2 Web & Mobile Apps Dev	Frontend Dev (Man-hours)	8,000 Hours	2,000 / hr	16,000,000
1.4.1 e-Nabız & HL7 Integration	Integration Engineers	9,000 Hours	2,000 / hr	18,000,000
1.6.1 On-site Staff Training	Clinical Trainers	25 hrs x 200 sessions	3,000 / hr	15,000,000

2.2 Three-Point Cost Estimation (PERT Formula: Expected Cost = (a + 4m + b) / 6)

WBS Task Name	Optimistic Cost (a)	Most Likely Cost (m)	Pessimistic Cost (b)	Expected PERT Cost (TRY)
1.2.1 Local Cloud Datacenter	10,000,000	15,000,000	25,000,000	15,833,333
1.2.2 Tablet Fleet Procurement	80,000,000	90,000,000	110,000,000	91,666,667
1.3.1 Central Database Dev	15,000,000	20,000,000	30,000,000	20,833,333
1.3.2 Web & Mobile Apps Dev	12,000,000	16,000,000	22,000,000	16,333,333

WBS Task Name	Optimistic Cost (a)	Most Likely Cost (m)	Pessimistic Cost (b)	Expected PERT Cost (TRY)
1.4.1 e-Nabız & HL7 Integration	15,000,000	18,000,000	25,000,000	18,666,667
1.6.1 On-site Staff Training	10,000,000	15,000,000	25,000,000	15,833,333